



Photon Protocol: Zero-Latency Consensus Pipelining

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Abstract

We present the Photon protocol, a pipelined binary consensus engine that achieves *zero-latency overhead* by overlapping sampling rounds for multiple transactions. In standard metastable consensus, each transaction occupies β sequential rounds before finalization, creating a throughput bottleneck when transactions arrive faster than the per-transaction finality time. Photon eliminates this bottleneck by pipelining: transaction t_{i+1} 's sampling begins in the same round as t_i 's, with each round's query carrying preferences for all in-flight transactions simultaneously. We formalize the pipeline as a *confidence matrix* $\mathbf{D} \in \mathbb{N}^{m \times d}$ tracking confidence across m concurrent transactions and d colors. We prove that pipelined sampling preserves the safety guarantee of sequential sampling: the probability of conflicting finalization remains $\leq 2^{-128}$ per transaction, with the total system safety degrading at most linearly in the number of concurrent transactions (bounded by the pipeline depth m). We analyze the pipeline stall conditions—configurations where one transaction's unresolved conflict blocks the finalization of dependent transactions—and prove that stalls are resolved in expected $O(1)$ additional rounds under honest supermajority. The protocol achieves *sustained throughput*: the finalization rate equals the transaction arrival rate for any arrival rate $T \leq n \cdot k / \Delta$ where Δ is the round time, regardless of the per-transaction finality latency. Empirical evaluation on a 1,000-node testnet demonstrates sustained 10,000 TPS with the same per-transaction latency as non-pipelined protocols (500ms), where non-pipelined throughput is limited to $1/\beta \approx 90$ transactions per second per conflict set.

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1 Introduction

In standard metastable consensus, a validator processes one transaction at a time per conflict set: it samples, waits for responses, updates confidence, and repeats for β rounds. This creates a throughput ceiling of $1/(\beta \cdot \Delta)$ finalizations per second per conflict set, where Δ is the round time. For $\beta = 11$ and $\Delta = 50\text{ms}$, this is merely 1.8 finalizations per second per conflict set.

While the Nova protocol amortizes this cost across DAG subgraphs, independent conflict sets still bottleneck at the per-conflict finalization rate. Photon addresses this by pipelining: multiple transactions enter the sampling pipeline simultaneously, sharing each round’s query-response cycle across all in-flight transactions.

The key challenge is ensuring that pipelining does not compromise safety. When multiple transactions are sampled in the same round, their confidence counters may interact (e.g., via shared Byzantine responses). We prove that this interaction is benign: the confidence dynamics for independent conflict sets are statistically independent, and for transactions within the same conflict set, pipelining reduces to sequential processing of the conflict resolution.

Contributions.

1. The Photon pipelined consensus protocol with confidence matrix formalization (§3).
2. A proof that pipelined safety degrades at most linearly in pipeline depth (§4).
3. Pipeline stall analysis and resolution bounds (§5).
4. Throughput analysis showing sustained finalization at the transaction arrival rate (§6).
5. Empirical evaluation achieving 10,000 TPS (§7).

2 System Model

Definition 2.1 (Pipeline). *A consensus pipeline of depth m processes m transactions simultaneously. Each round, the validator’s query carries preferences for all m transactions, and the response includes preferences for all m transactions.*

Definition 2.2 (Confidence Matrix). *The confidence matrix $\mathbf{D} \in \mathbb{N}^{m \times d}$ has entry $D_{i,c}$ representing the cumulative confidence count for transaction i and color c . Transaction i is finalized when $\max_c D_{i,c} \geq \beta$.*

Definition 2.3 (Pipeline Slot). *A pipeline slot $s \in \{1, \dots, m\}$ is occupied by a transaction from the time it enters the pipeline until it is finalized (and the slot is freed for a new transaction).*

3 The Photon Protocol

3.1 Pipelined Sampling

Algorithm 1 Photon Protocol — Pipelined Consensus

Require: Validator v , pipeline depth m

Require: Parameters: k (sample size), α (quorum), β (threshold)

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1: pipeline[]  $\leftarrow$  array of  $m$  slots, initially empty
2: loop
3:                                      $\triangleright$  Fill empty slots with pending transactions
4:   for each empty slot  $s$  do
5:     if  $\exists$  pending transaction  $t_s$  then
6:       pipeline[ $s$ ]  $\leftarrow t_s$ ;  $D_{s,c} \leftarrow 0$  for all  $c$ 
7:     end if
8:   end for
9:    $S \leftarrow \text{RandomSample}(V \setminus \{v\}, k)$ 
10:                                      $\triangleright$  Batch query: ask for all pipeline transactions at once
11:   Query each  $u \in S$  for  $\{\text{col}(u, t_s) : s \in \text{active slots}\}$ 
12:   for each active slot  $s$  with transaction  $t_s$  do
13:     for each color  $c$  do
14:        $n_c \leftarrow |\{u \in S : \text{col}(u, t_s) = c\}|$ 
15:       if  $n_c \geq \alpha k$  then
16:          $D_{s,c} \leftarrow D_{s,c} + 1$ 
17:       end if
18:     end for
19:      $\text{col}(v, t_s) \leftarrow \arg \max_c D_{s,c}$ 
20:     if  $\max_c D_{s,c} \geq \beta$  then
21:       finalize  $t_s$  with color  $\arg \max_c D_{s,c}$ 
22:       Free slot  $s$ 
23:     end if
24:   end for
25: end loop

```

3.2 Message Format

Each query message carries a vector of m transaction identifiers, and each response carries a vector of m color preferences. The message overhead compared to single-transaction queries is $O(m)$ per message, resulting in amortized $O(1)$ per transaction per round.

For $m = 100$ transactions and 32-byte transaction IDs + 1-byte color each, the query overhead is $100 \times 33 = 3.3\text{KB}$, well within typical UDP datagram limits.

4 Safety Analysis

4.1 Independence of Confidence Counters

Theorem 4.1 (Pipelined Safety). *For m independent transactions (no shared conflict sets) in the Photon pipeline, the probability that any transaction experiences a safety violation is at most:*

$$\Pr[\text{any violation}] \leq m \cdot (q_0/q_1)^\beta$$

where q_0, q_1 are the supermajority probabilities for minority and majority colors.

Proof. Consider two transactions t_i, t_j in the pipeline with independent conflict sets. Their confidence counters $D_{i,\cdot}$ and $D_{j,\cdot}$ evolve based on the same sample S but independent color queries.

For transaction t_i , the responses from S about t_i 's conflict are drawn from the population distribution for t_i 's conflict set. Similarly for t_j . Since the conflict sets are disjoint, the responses are independent random variables (the same validator may respond differently for t_i and t_j).

Therefore, the confidence counter evolution for t_i is statistically independent of t_j 's. The safety violation probability for each transaction is $(q_0/q_1)^\beta$ (from the Wave analysis), and by a union bound over m transactions:

$$\Pr[\text{any violation}] \leq m \cdot (q_0/q_1)^\beta$$

For $m = 100$ and $(q_0/q_1)^\beta < 2^{-128}$: $100 \cdot 2^{-128} < 2^{-121}$, still negligible. $\square$

4.2 Conflicting Pipeline Entries

Lemma 4.2 (Same-Conflict Serialization). *If two transactions t_i, t_j in the pipeline belong to the same conflict set $\mathcal{C}$, the Photon protocol serializes their finalization: t_j cannot finalize until t_i 's conflict is resolved.*

Proof. When $t_i \in \mathcal{C}(t_j)$, the validator's preference for t_j depends on t_i 's outcome. The confidence counter for t_j cannot reach β until the population has converged on t_i 's resolution, because the supermajority condition for t_j requires consistency with t_i 's resolution. This serialization adds at most β additional rounds for the second transaction, which is the normal single-transaction finality time. $\square$

5 Pipeline Stall Analysis

5.1 Stall Definition

Definition 5.1 (Pipeline Stall). *A stall occurs when a pipeline slot s fails to increment its leading confidence counter for τ consecutive rounds, blocking the slot from finalization and preventing new transactions from entering.*

Theorem 5.1 (Stall Duration). *The expected stall duration (number of consecutive non-incrementing rounds) for a pipeline slot is:*

$$\mathbb{E}[\tau] = \frac{1}{q_1} + O\left(\frac{q_0}{q_1^2}\right)$$

which is $O(1)$ when $q_1 > 1/2$.

Proof. A non-incrementing round occurs when the sample does not produce a supermajority for the leading color. After population convergence (which takes $O(\log n)$ rounds), the probability of a supermajority round is $q_1 \approx \Phi(p, k, \alpha)$ where $p > 1 - \gamma$.

The number of rounds until the next supermajority round follows a geometric distribution with parameter q_1 , giving $\mathbb{E}[\tau] = 1/q_1$. For $q_1 = 0.63$ (standard parameters with $p = 0.8$, $k = 20$, $\alpha = 0.8$), the expected stall is 1.6 rounds. $\square$

Corollary 5.2 (Pipeline Utilization). *The expected pipeline utilization (fraction of rounds where at least one slot advances) is $1 - (1 - q_1)^m$, which exceeds 0.99 for $m \geq 7$ with standard parameters.*

Proof. A round is wasted only if none of the m active slots produce a supermajority. Since the slots' supermajority events are independent (for independent conflict sets), the probability of a fully idle round is $(1 - q_1)^m = (0.37)^m < 0.01$ for $m \geq 7$. $\square$

6 Throughput Analysis

Theorem 6.1 (Sustained Throughput). *The Photon protocol sustains finalization throughput of:*

$$\Theta_{\text{sustained}} = \min\left(T, \frac{m \cdot q_1}{\Delta}\right)$$

where T is the transaction arrival rate, m is the pipeline depth, q_1 is the per-round supermajority probability, and Δ is the round time.

Proof. Each of the m pipeline slots produces one finalization every $1/(q_1 \cdot \beta^{-1})$ rounds, where β rounds of supermajority are needed and each round succeeds with probability q_1 . The expected slot cycle time is β/q_1 rounds = $\beta\Delta/q_1$ seconds.

With m slots operating in parallel, the finalization rate is:

$$\Theta_{\text{sustained}} = \frac{m}{\beta\Delta/q_1} = \frac{mq_1}{\beta\Delta}$$

For $m = 100$, $q_1 = 0.63$, $\beta = 11$, $\Delta = 0.05\text{s}$: $\Theta_{\text{sustained}} = 100 \times 0.63 / (11 \times 0.05) \approx 114$ finalizations per round = $114/0.05 \approx 2,280$ per second per conflict-set pipeline.

Since most transactions have independent conflict sets, the total throughput is $\Theta_{\text{sustained}} \times C$ where C is the number of active conflict sets, easily reaching 10,000+ TPS. $\square$

7 Empirical Evaluation

7.1 Setup

1,000-node testnet, pipeline depth $m = 128$, parameters $k = 20$, $\alpha = 0.8$, $\beta = 11$, round time $\Delta = 50\text{ms}$.

7.2 Results

Protocol	TPS	Latency (median)	Messages/tx
Sequential Wave	90/conflict	550ms	220
Nova (burst)	8,500	400ms	2.2
Photon (pipelined)	10,000	500ms	1.8

Table 1: Throughput comparison across consensus protocols

Photon achieved 10,000 TPS—the transaction arrival rate—with 500ms median per-transaction latency (same as non-pipelined). The key advantage is that throughput is decoupled from per-transaction latency: increasing β for stronger safety does not reduce throughput.

8 Conclusion

Photon introduces consensus pipelining to the metastable protocol family, enabling sustained throughput at the transaction arrival rate independent of per-transaction finality latency. The confidence matrix formalization ensures that pipelined safety degrades at most linearly in the pipeline depth, and the stall analysis shows that pipeline utilization exceeds 99% for depths ≥ 7 . Photon is the binary consensus engine in the Lux Quasar stack, providing the per-conflict sampling primitive that underlies the Nova burst finalization and Prism parallel consensus mechanisms.

References

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